Theory Behind the Simulation

2D Explicit FEA — Copper Panel Impact | explains the method implemented in copper_panel_impact_fea.m

1. Problem statement

A steel sphere of mass m_d travels at v_0 = 1000 m/s toward a thin elastic plate. On contact the plate is loaded locally, a stress wave spreads outward at roughly the material wave speed, and the plate flexes and rebounds. The goal is to follow this transient in time and record how far the plate moves.

Two plates are analysed independently:

- a **copper C10100** strip (280 mm × 20 mm, 1.32 mm thick), clamped at one end;
- the **top wall of a Ti-6Al-4V frame** (200 mm × 10 mm, 10 mm thick), clamped at both ends.

Because the event lasts microseconds and involves wave propagation, an **explicit dynamic** method is the natural choice. Implicit methods need a global equation solve at every step, while explicit methods only need cheap element-level force evaluations.

2. Plane-stress elasticity

Each plate is treated as a thin 2D sheet loaded in its own plane, so stresses through the thickness are neglected ($\sigma_z = \tau_{xz} = \tau_{vz} = 0$). Stress and strain are stored in Voigt form:

$$\sigma = [\, \sigma_x,\ \sigma_v,\ \tau_{xv} \,]^T$$

$$\varepsilon = [\, \varepsilon_x,\ \varepsilon_y,\ \gamma_{xy} \,]^T = [\, \partial u/\partial x,\ \partial v/\partial y,\ \partial u/\partial y + \partial v/\partial x \,]^T$$

Hooke's law for plane stress is $\sigma = D\,\varepsilon$ with

$$D = E / (1 - \nu^2) \times \begin{vmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu)/2 \end{vmatrix}$$

This is exactly Dm_cu and Dm_ti in the code. The material is **linear elastic**; no yield, hardening or failure is modelled.

3. Finite element discretisation (Q4)

3.1 Mesh

Each plate is a structured grid of 4-node bilinear quadrilaterals. With NX × NY elements there are (NX+1)(NY+1) nodes, each with two degrees of freedom (u, v).

Mesh	Elements	Nodes	Element size
Copper panel	80 × 10 = 800	81 × 11 = 891	3.5 mm × 2 mm
Titanium wall	50 × 5 = 250	51 × 6 = 306	4 mm × 2 mm

Nodes in each element are numbered counter-clockwise from the lower-left corner: [n1, n1+1, n1+NX+2, n1+NX+1].

3.2 Shape functions

On the parent element with natural coordinates (ξ, η) in [−1, 1]:

$$N_i(\xi, \eta) = \tfrac{1}{4}\,(1 + \xi\,\xi_i)(1 + \eta\,\eta_i),\ i = 1 \ldots 4$$

3.3 Strain–displacement matrix at the element centre

For a rectangular element of size $e_x \times e_y$ the Jacobian is constant: $\partial x/\partial \xi = e_x/2$ and $\partial y/\partial \eta = e_y/2$. Evaluating the derivatives at the centre $(\xi, \eta) = (0, 0)$:

$$\partial N_i/\partial x = \xi_i / (2\, e_x),\ \partial N_i/\partial y = \eta_i / (2\, e_y)$$

which gives, for the four nodes,

$$\partial N/\partial x = (1 / 2e_x)\ [\,-1\ 1\ 1\ -1\,],\ \partial N/\partial y = (1 / 2e_y)\ [\,-1\ -1\ 1\ 1\,]$$

These fill the 3×8 matrix B (Bp, Bf in the code), with $N_{i,x} = \partial N_i/\partial x$ and $N_{i,y} = \partial N_i/\partial y$:

$$\begin{vmatrix} N_{1,x} & 0 & N_{2,x} & 0 & N_{3,x} & 0 & N_{4,x} & 0 \\ 0 & N_{1,y} & 0 & N_{2,y} & 0 & N_{3,y} & 0 & N_{4,y} \\ N_{1,y} & N_{1,x} & N_{2,y} & N_{2,x} & N_{3,y} & N_{3,x} & N_{4,y} & N_{4,x} \end{vmatrix}$$

so that the element strain is $\varepsilon_e = B\, u_e$, where $u_e = [u_1, v_1, \ldots, u_4, v_4]^T$.

3.4 One-point integration

Because B is evaluated only at the element centre, the element uses **single-point (reduced) integration**. For a rectangle the element volume is

$$V_e = e_x \cdot e_y \cdot t$$

where t is the plate thickness (Ve_p, Ve_f in the code). The internal nodal force vector of one element is

$$f_e^{\,int} = B^T \sigma\, V_e$$

Reduced integration is cheap and avoids shear locking, but it leaves **hourglass modes** (deformation patterns that produce zero strain at the centre) uncontrolled. Coarse meshes can therefore show spurious, non-physical oscillations. Production explicit codes add hourglass stiffness or viscosity to suppress them.

3.5 Assembly

Element forces are scattered into the global vectors f^{int} (fi_p, fi_f) by adding each element's contribution to its four nodes.

4. Lumped mass matrix

The consistent mass matrix is replaced by a diagonal (lumped) one, which makes the explicit update trivially cheap. Each element distributes its mass equally to its four nodes:

$$m_{node} \mathrel{+}= \rho\, t\, e_x\, e_y / 4$$

The code stores ρt as an areal density (rho2d_cu, rho2d_f) and accumulates into the vectors Mp and Mf. A diagonal M means the equations of motion decouple node by node.

5. Explicit time integration

The semi-discrete equation of motion is

$$M\, \ddot{u} = f^{ext} - f^{int}(u)$$

where f^{ext} is only the contact force from the debris. Each step computes the acceleration of every free node,

$$a^n = M^{-1}\ (\, f^{c,n} - f^{int,n}\,)$$

then advances velocity first and displacement second:

$$v^{n+1} = v^n + \Delta t \cdot a^n,\ u^{n+1} = u^n + \Delta t \cdot v^{n+1}$$

This is the **semi-implicit (symplectic) Euler** scheme. It is algebraically equivalent to the **leapfrog / central-difference** method with velocities stored at half steps, the standard scheme in explicit codes such as LS-DYNA and Abaqus/Explicit. It is second-order accurate for the undamped case and conditionally stable. The source comment calls it "Velocity-Verlet"; strictly it is the leapfrog form.

No Rayleigh or bulk viscosity is included, so there is no numerical damping of high-frequency noise.

6. Stable time step (CFL condition)

An explicit scheme is only stable if the time step is smaller than the time a stress wave needs to cross the smallest element. The 1D elastic bar wave speed is

$$c = \sqrt{(E / \rho)}$$

and the time step is

$$\Delta t = \alpha \cdot L_{min} / c_{max},\ \alpha = 0.80$$

L_{min} is the smallest element edge over both meshes, and c_{max} is the larger of the copper and titanium wave speeds. Using one global Δt for both meshes is simple and conservative (copper could take a larger step on its own).

The in-plane dilatational wave speed under plane stress is slightly higher than the bar speed:

$$c_p = \sqrt{(E / (\rho (1 - \nu^2)))}$$

The safety factor α = 0.8 leaves margin for this and for the mesh's actual stability limit, which for 2D quads is somewhat stricter than the simple 1D estimate. If displacements ever blow up, reduce α.

7. Penalty contact

The debris is a circle of radius R_d centred at (x_d, y_d), moving as a rigid body. A plate node at deformed position $x_i = X_i + u_i$ is in contact when its distance to the centre,

$$d_i = | x_i - x_d |,$$

is less than R_d. The overlap is $\delta_i = R_d - d_i$, and a linear spring pushes the node outward along the line of centres:

$$f_i^c = k \, \delta_i \, n_i,\ n_i = (x_i - x_d) / d_i$$

By Newton's third law the debris receives the opposite force, so its velocity changes by

$$\Delta v_d = -(\Delta t / m_d) \, \Sigma_i \, f_i^c$$

and its position is advanced by $x_d \mathrel{+}= v_d \, \Delta t$. The penalty stiffness is set to

$$k = 32 \cdot E \cdot R_d$$

which has units of N/m (Pa × m). It is a numerical parameter: too soft lets the debris sink into the plate, too stiff forces a smaller stable time step. The debris mass is that of a sphere,

$$m_d = (4/3) \, \pi \, R_d^{\,3} \, \rho_d$$

while the contact geometry is a 2D disk, a pragmatic simplification of the true 3D problem. Contact detection stops (done1, done2) once the debris has passed well below the plate.

8. Von Mises stress

Stress is shown as the von Mises equivalent stress for plane stress:

$$\sigma_{vM} = \sqrt{(\sigma_x^{\,2} - \sigma_x \sigma_y + \sigma_y^{\,2} + 3 \, \tau_{xy}^{\,2})}$$

It measures how close the local stress state is to yielding in ductile metals. In this code it is used only for **colouring**: each frame divides by the current maximum, so the colour bar is relative. Comparing it with a yield strength (annealed copper is in the tens of MPa, Ti-6Al-4V near 900 MPa) is a useful post-check, but the model itself never yields.

9. Boundary conditions

Fixed nodes are enforced three ways at once: acceleration set to zero; displacement and velocity reset to zero after every update; and nodal mass multiplied by 200 (extra insurance against numerical motion). The fixed nodes are:

- **Copper panel:** the left edge, $x = x_0$ (cantilever-style clamp).
- **Titanium wall:** the left and right edges ($x = \pm 100$ mm), a fixed–fixed span.

10. Worked numbers for this model

Quantity	Formula	Value
Copper wave speed	$\sqrt{(E_{cu}/\rho_{cu})}$	≈ 3583 m/s
Titanium wave speed	$\sqrt{(E_{ti}/\rho_{ti})}$	≈ 5117 m/s
Smallest element edge	$\min(e_x, e_y)$ over both meshes	2 mm
Time step	$0.8\ L_{min} / c_{max}$	≈ 312.7 ns
Number of steps	ceil(650 µs / Δt)	2079
Debris mass	$(4/3)\ \pi\ R_d^3\ \rho_d$	≈ 0.514 g
Debris kinetic energy	$½\ m_d\ v_0^2$	≈ 257 J
Time to first contact	0.002 m / 1000 m/s	≈ 2 µs
Penalty stiffness (Cu / Ti)	$32\ E\ R_d$	≈ 9.2e9 / 9.3e9 N/m

At 312 ns per step the 650 µs simulation spans about 2000 steps. In that time a wave in copper travels roughly 2.3 m, so it crosses the 280 mm panel and reflects from the clamped end several times. That is why the deflection history shows oscillation rather than a single peak.

11. Validity and sanity checks

Things worth checking if you extend or rely on this model:

1. **Energy balance.** Total energy (debris kinetic + plate kinetic + strain energy) should be conserved after contact, apart from small penalty-contact errors. A growing total energy signals instability.
2. **Momentum.** Debris momentum lost should match the impulse delivered to the plate plus the reaction at the clamp.
3. **Mesh convergence.** Refine NX_* / NY_* and confirm the peak deflection changes only slightly. With one-point integration and no hourglass control, coarse meshes are the main risk.
4. **Penalty sensitivity.** Vary the factor 32 in k_pen_* by ×0.5 and ×2; results should not change much.
5. **Plasticity.** A 257 J, 1 km/s strike on a 1.32 mm copper strip would exceed copper's elastic range locally. For quantitative damage predictions use an elastic–plastic model with strain-rate dependence (for example Johnson–Cook) and an erosion or failure criterion, ideally in a validated explicit code.

12. References

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